



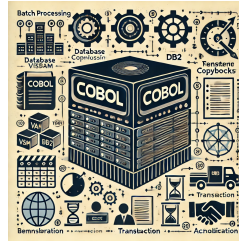
Architecture Model

CLBS-A-001

16 September 2026

Investment Portfolio Management System Architecture Model

Document Responsibility: COBOL Legacy Benchmark Suite Maintainers



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1 Scope

This Architecture Model document describes the as-built structure of the COBOL Legacy Benchmark Suite (CLBS), an Investment Portfolio Management System implemented in Enterprise COBOL for z/OS. It identifies the architectural layers of the system, the programs and copybooks allocated to each layer, the data stores they own, and the interfaces between them.

The model is extracted from the source tree itself - COBOL programs, JCL procedures, CICS resource definitions, VSAM cluster definitions and DB2 DDL - and is intended as the reference model for modernization, translation and decomposition work performed against this repository.

Commentary Note:

The system is a benchmark artifact. It is written to production standards for z/OS, but it is not deployed; the model therefore describes intended runtime structure rather than an operating installation.

2 Conflicts and Deviations

- 2.1 Where this model conflicts with the source code in src/, the source code governs and this document shall be corrected.
- 2.2 Where this model conflicts with documentation/technical/system-architecture.md, the discrepancy shall be recorded in the Revision Summary and resolved in favour of the source code.
- 2.3 Components listed as planned or not yet implemented shall not be treated as available interfaces by translation or modernization tooling.

Commentary Note:

Two such deviations are known and recorded in this issue: the batch position update program (POSUPDT) is a placeholder file with no PROGRAM-ID, and the transaction validation program TRNVAL00 referenced by the architecture narrative is not present in src/programs.

3 References

Project References

documentation/technical/system-architecture.md

System Architecture Document, v1.2

documentation/technical/data-dictionary.md

Data Dictionary

documentation/technical/development-backlog.md

Development Backlog

documentation/operations/test-data-specs.md

Test Data Specifications

src/cics/PORTDFN.csd *CICS Resource Definitions, group PORTGRP*

src/database/db2/db2-definitions.sql

DB2 Table and Index Definitions

src/database/vsam/vsam-definitions.txt

VSAM Cluster Definitions

Platform References

Enterprise COBOL for z/OS

Language Reference and Programming Guide

CICS Transaction Server for z/OS

Application Programming Reference

Db2 for z/OS

SQL Reference and Application Programming Guide

z/OS MVS JCL Reference *Job Control Language Reference*

4 General Requirements

The architecture of the system shall satisfy the following.

- 4.1 Programs shall be allocated to exactly one architectural layer - batch, online, portfolio maintenance, common services, utility or test - matching their directory under src/programs.
- 4.2 Record layouts shared between programs shall be held in copybooks under src/copybook and shall not be redeclared in program source.
- 4.3 Online programs shall be pseudo-conversational and shall exchange state through the COMMAREA layouts defined in src/copybook/online.

- 4.4 Batch programs shall record execution state in the batch control file and shall support checkpoint and restart at sync points.
- 4.5 DB2 access from batch shall be routed through the common DB2 service programs (DB2CONN, DB2CMT, DB2ERR, DB2STAT); online DB2 access shall be routed through DB2ONLN, CURSMGR and DB2RECV.
- 4.6 Error handling shall be centralized in ERRPROC for batch and non-CICS callers and in ERRHNDL for CICS programs; both shall persist diagnostics to the DB2 ERRLOG table or the error log file.
- 4.7 All online access shall be authorized by SECMGR against the authorization file, with security events written to the audit log.
- 4.8 VSAM shall be used as the primary store for master and position records; DB2 shall be used for long-term history, return codes and error logging.

Commentary Note:

Requirement 4.5 is the principal decomposition seam for modernization: the DB2 service programs are the only components that own connection, commit and cursor lifecycle.

5 Architecture Model

- 5.1 Layering. The system is organized into six application layers over two data platforms. The batch layer executes scheduled workloads under JCL; the online layer executes under CICS transaction PINQ; the portfolio layer provides master-file CRUD and transaction posting; the common services layer provides DB2, error and audit services to the other layers; the utility and test layers provide operational and benchmark support. Layer membership, component counts and dependencies are listed in Appendix-1.
- 5.2 Batch layer. BCHCTL00 and PRCSEQ00 drive process control and sequencing against the batch control and process sequence files; CKPRST provides checkpoint and restart services; RCVPRC00 performs recovery processing; HISTLD00 loads transaction history from VSAM into the DB2 POSHIST table; RPTPOS00, RPTAUD00 and RPTSTA00 produce position, audit and statistics reports; RTNCDE00 and RTNANA00 record and analyse program return codes in the DB2 RTNCODES table.
- 5.3 Online layer. CICS transaction PINQ starts INQONLN, the inquiry controller, which links to SECMGR for authorization, INQPORT for portfolio and position inquiry, INQHIST for transaction history, and ERRHNDL for error presentation.

INQHIST links to CURSMGR for cursor lifecycle and to DB2ONLN and DB2RECV for DB2 access and recovery. Screens are defined by mapset INQSET; the position file POSFILE is defined read-only with browse enabled.

- 5.4 Portfolio layer. PORTMSTR is the master maintenance module over the portfolio VSAM KSDS, with PORTADD, PORTUPDT, PORTDEL and PORTREAD providing the individual batch-driven operations, PORTVALD providing field validation and PORTTRAN posting transactions against portfolio records. PORTMSTR and PORTTRAN call AUDPROC and ERRPROC so that every mutation is audited.
- 5.5 Common services layer. DB2CONN establishes and retries DB2 connections, DB2CMT owns commit and rollback scope, DB2ERR classifies SQLCODEs and persists them to ERRLOG, DB2STAT captures access statistics, ERRPROC is the central error handler and AUDPROC writes the audit trail. These programs are called, never linked, and carry no CICS dependency.
- 5.6 Utility and test layers. UTLMNT00 performs file maintenance, archiving and reorganization; UTLMON00 collects performance metrics and raises alerts; UTLVAL00 validates and reconciles position and history data. TSTGEN00 generates benchmark portfolios and transactions and TSTVAL00 executes test cases and compares expected against actual results.
- 5.7 Data architecture. VSAM key-sequenced data sets hold the portfolio master, position master and transaction history records; sequential data sets carry transaction input, reports, alerts and archives; DB2 database POSMVP holds the POSHIST history table, the ERRLOG error table and the RTNCODES return-code table, together with the relational PORTFOLIO_MASTER, INVESTMENT_POSITIONS and TRANSACTION_HISTORY tables. Data stores and their owning components are listed in Appendix-2.
- 5.8 Interfaces. Three interface mechanisms exist: static COBOL CALL for batch and common services, EXEC CICS LINK with a COMMAREA for online programs, and file or table access for data-mediated coupling between batch steps. The complete relationship export is listed in Appendix-3.

Commentary Note:

There is no synchronous interface between the online and batch layers. Coupling is exclusively through shared VSAM files and DB2 tables, which is what makes the layers separable in a modernization effort.

6 Roles and Responsibilities

Maintenance of this model is the responsibility of the repository maintainers. The model shall be reissued when a program, copybook, JCL job, CICS resource or data store is added, removed or renamed, and when a layer allocation changes. Appendix-5 records the responsibility matrix.

Revision Summary

16 September 2026	First issue. Model extracted from the source tree at repository head; documents the six application layers, 40 components, 13 data stores, 15 batch jobs and one CICS transaction.
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Appendix-1

Component Inventory Matrix

Component	Layer	Responsibility	Copybooks	Data stores
BCHCTL00	Batch Processing	Batch control and process status manager	BCHCON, BCHCTL, ERRHAND	BATCH-CONTROL-FILE
PRCSEQ00	Batch Processing	Process sequencing and dependency evaluation	BCHCON, BCHCTL, ERRHAND, PRCSEQ	BATCH-CONTROL-FILE, PROCESS-SEQ-FILE
RCVPRC00	Batch Processing	Recovery processing for failed batch processes	BCHCON, BCHCTL, ERRHAND, PRCSEQ	BATCH-CONTROL-FILE, PROCESS-SEQ-FILE
CKPRST	Batch Processing	Checkpoint and restart services	CKPRST, RETHND	CHECKPOINT-FILE
HISTLD00	Batch Processing	Loads VSAM transaction history into DB2 POSHIST	BCHCON, BCHCTL, DBPROC, DBTBLS, ERRHAND, HISTREC, SQLCA	BATCH-CONTROL-FILE, TRANSACTION-HISTORY, POSHIST
RPTPOS00	Batch Processing	Daily position and valuation report	ERRHAND, POSREC, RTNCODE, TRNREC	POSITION-MASTER, TRANSACTION-HISTORY, REPORT-FILE
RPTAUD00	Batch Processing	Audit and exception report	AUDITLOG, ERRHAND, RTNCODE	AUDIT-FILE, ERROR-FILE, REPORT-FILE
RPTSTA00	Batch Processing	System statistics report	BCHCTL, DB2STAT, ERRHAND, RTNCODE	BATCH-STATS, DB2-STATS, REPORT-FILE
RTNCDE00	Batch Processing	Records program return codes in DB2	RTNCODE	RTNCODES
RTNANA00	Batch Processing	Return code trend analysis report	-	RTNCODES, REPORT-FILE
POSUPDT	Batch Processing	Position update engine [not implemented]	-	-
INQONLN	Online Inquiry	Inquiry controller and screen flow	ERRHND, INQCOM	-
INQPORT	Online Inquiry	Portfolio and position inquiry	INQCOM, POSREC	POSFILE
INQHIST	Online Inquiry	Transaction history inquiry	INQCOM	POSHIST
CURSMGR	Online Inquiry	DB2 cursor lifecycle management	-	POSHIST
SECMGR	Online Inquiry	Authorization and security event logging	ERRHND	AUTHFILE, AUDITLOG
ERRHNDL	Online Inquiry	Online error handling and message formatting	ERRHND	ERRLOG
DB2ONLN	Online Inquiry	Online DB2 access controller	ERRHND	-
DB2RECV	Online Inquiry	Online DB2 error recovery	DB2REQ, ERRHND	-
PORTMSTR	Portfolio Maintenance	Portfolio master maintenance (VSAM CRUD)	-	PORTFOLIO-FILE
PORTADD	Portfolio Maintenance	Add portfolio records from input file	PORTFLIO	PORTFOLIO-FILE, INPUT-FILE
PORTUPDT	Portfolio Maintenance	Apply portfolio updates	PORTFLIO	PORTFOLIO-FILE, UPDATE-FILE
PORTDEL	Portfolio Maintenance	Delete portfolio records with audit trail	PORTFLIO	PORTFOLIO-FILE, DELETE-FILE, AUDIT-FILE

Component	Layer	Responsibility	Copybooks	Data stores
PORTREAD	Portfolio Maintenance	Read and report portfolio records	PORTFLIO	PORTFOLIO-FILE
PORTVALD	Portfolio Maintenance	Portfolio field validation	PORTVAL	-
PORTTRAN	Portfolio Maintenance	Posts transactions against portfolio records	AUDITLOG, ERRHAND, PORTREC, TRNREC	PORTFOLIO-FILE, TRANSACTION-FILE
PORTTEST	Portfolio Maintenance	Portfolio test data generation	ERRHAND, PORTFLIO	TEST-FILE
DB2CONN	Common Services	DB2 connection establishment and retry	DBPROC, ERRHAND, SQLCA	-
DB2CMT	Common Services	Commit and rollback scope control	DBPROC, ERRHAND, SQLCA	-
DB2ERR	Common Services	SQLCODE classification and ERRLOG persistence	DBPROC, DBTBLS, ERRHAND, SQLCA	ERRLOG
DB2STAT	Common Services	DB2 access statistics collection	DBPROC, ERRHAND, SQLCA	-
ERRPROC	Common Services	Central error handler for non-CICS callers	ERRHAND	ERROR-LOG
AUDPROC	Common Services	Audit trail writer	AUDITLOG	AUDIT-FILE
UTLMNT00	Utility	File maintenance	ERRHAND, RTNCODE	CONTROL-FILE, ARCHIVE-FILE, REPORT-FILE
UTLMON00	Utility	Resource monitoring and alerting	DB2STAT, ERRHAND, RTNCODE	MONITOR-CONFIG, MONITOR-LOG, ALERT-FILE, DB2-STATS
UTLVAL00	Utility	Data integrity validation and reconciliation	ERRHAND, POSREC, RTNCODE, TRNREC	POSITION-MASTER, TRANSACTION-HISTORY, VALIDATION-CONTROL, ERROR-REPORT
TSTGEN00	Test and Benchmark	Benchmark portfolio and transaction generation	ERRHAND, PORTFLIO, RTNCODE, TRNREC	TEST-CONFIG, PORTFOLIO-OUT, TRANSACTION-OUT
TSTVAL00	Test and Benchmark	Test case execution and result comparison	ERRHAND, RTNCODE	TEST-CASES, EXPECTED-RESULTS, ACTUAL-RESULTS, TEST-REPORT

Appendix-2

Data Stores, Jobs and Transactions

Data store	Platform	Data set / table	Key	Owner
PORTFOLIO-FILE	VSAM KSDS, 400 bytes	PORTFOLIO.MASTER.FILE	12 bytes at offset 0	PORTMSTR
POSITION-MASTER	VSAM KSDS, 200 bytes	PORTFOLIO.POSITION.VSAM	Portfolio and security identifier	POSUPDT
TRANSACTION-HISTORY	VSAM KSDS, variable	PORTFOLIO.TRANHIST	Portfolio and timestamp	HISTLD00
BATCH-CONTROL-FILE	VSAM KSDS, BCHCTL layout	PORTFOLIO.BCHCTL	Process identifier	BCHCTL00
PROCESS-SEQ-FILE	VSAM KSDS, PRCSEQ layout	PORTFOLIO.PRCSEQ	Sequence number	PRCSEQ00
CHECKPOINT-FILE	VSAM, CKPRST layout	PORTFOLIO.CHKPT	Process identifier	CKPRST
AUDIT-FILE	Sequential, AUDITLOG layout	PORTFOLIO.AUDIT	n/a	AUDPROC
ERROR-LOG	Sequential, ERRHAND layout	PORTFOLIO.ERRLOG	n/a	ERRPROC
POSHIST	DB2 table (POSMVP), relational	POSHIST	PORTFOLIO_ID, TRANS_DATE	HISTLD00
ERRLOG	DB2 table, relational	ERRLOG	ERROR_TIMESTAMP, PROGRAM_ID	DB2ERR
RTNCODES	DB2 table, relational	RTNCODES	PROGRAM_ID, RETURN_CODE	RTNCDE00
POSFILE	CICS FILE over VSAM, 200 bytes	PORTFOLIO.POSITION.VSAM	Position key	INQPORT
AUTHFILE	VSAM, auth layout	PORTFOLIO.AUTH	User identifier	SECMGR

Job	Program	JCL	Purpose
RPTPOS	RPTPOS00	src/jcl/batch/RPTPOS.jcl	Daily position report
RPTAUD	RPTAUD00	src/jcl/batch/RPTAUD.jcl	Audit report
RPTSTA	RPTSTA00	src/jcl/batch/RPTSTA.jcl	System statistics report
RTNANA	RTNANA00	src/jcl/RTNANA.jcl	Return code analysis
PORTDEF	IDCAMS	src/jcl/portfolio/PORTDEF.jcl	Define portfolio VSAM cluster
PORTADD	PORTADD	src/jcl/portfolio/PORTADD.jcl	Add portfolios
PORTUPDT	PORTUPDT	src/jcl/portfolio/PORTUPDT.jcl	Update portfolios
PORTDEL	PORTDEL	src/jcl/portfolio/PORTDEL.jcl	Delete portfolios
PORTREAD	PORTREAD	src/jcl/portfolio/PORTREAD.jcl	Read portfolios
PORTTEST	PORTTEST	src/jcl/portfolio/PORTTEST.jcl	Generate portfolio test data
UTLMNT	UTLMNT00	src/jcl/utility/UTLMNT.jcl	File maintenance
UTLMON	UTLMON00	src/jcl/utility/UTLMON.jcl	System monitoring
UTLVAL	UTLVAL00	src/jcl/utility/UTLVAL.jcl	Data validation
TSTGEN	TSTGEN00	src/jcl/test/TSTGEN.jcl	Test data generation

Job	Program	JCL	Purpose
TSTVAL	TSTVAL00	src/jcl/test/TSTVAL.jcl	Test validation

Tran	Program	Mapset	Group	DB2ENTRY	Plan	Purpose
PINQ	INQONLN	INQSET	PORTGRP	PORTDB2	PORTPLAN	Portfolio inquiry

Appendix-3

Interface and Dependency Matrix

Source	Interface	Target	Detail
z/OS Scheduler	submits	BCHCTL00	Batch job stream
BCHCTL00	call	ERRPROC	Error reporting
PRCSEQ00	call	ERRPROC	Error reporting
RCVPRC00	call	ERRPROC	Error reporting
HISTLD00	call	ERRPROC	Error reporting
HISTLD00	sql	POSHIST	INSERT history rows
HISTLD00	file	TRANSACTION-HISTORY	Sequential read of VSAM history
RPTPOS00	file	POSITION-MASTER	Read positions
RPTAUD00	file	AUDIT-FILE	Read audit trail
RPTSTA00	file	BATCH-CONTROL-FILE	Read process statistics
RTNCDE00	sql	RTNCODES	Insert return codes
RTNANA00	sql	RTNCODES	Query return codes
Portfolio Analyst	uses	PINQ	3270 terminal
PINQ	starts	INQONLN	CICS transaction definition
INQONLN	link	SECMGR	EXEC CICS LINK
INQONLN	link	INQPORT	EXEC CICS LINK
INQONLN	link	INQHIST	EXEC CICS LINK
INQONLN	link	ERRHNDL	EXEC CICS LINK
INQHIST	link	CURSMGR	Cursor open
INQHIST	link	DB2ONLN	DB2 access
INQHIST	link	DB2RECV	Recovery on SQL failure
DB2RECV	link	DB2ONLN	Connection reset
DB2RECV	link	ERRHNDL	Error reporting
INQPORT	file	POSFILE	CICS READ and BROWSE
INQHIST	sql	POSHIST	Cursor over history rows
SECMGR	file	AUTHFILE	Authorization lookup
SECMGR	file	AUDIT-FILE	Security event logging
ERRHNDL	sql	ERRLOG	Insert error rows
PORTMSTR	file	PORTFOLIO-FILE	VSAM CRUD
PORTMSTR	call	AUDPROC	Audit mutations
PORTMSTR	call	ERRPROC	Error reporting
PORTTRAN	call	AUDPROC	Audit postings
PORTTRAN	call	ERRPROC	Error reporting
PORTTRAN	file	PORTFOLIO-FILE	Update balances
PORTADD	file	PORTFOLIO-FILE	Insert records
PORTUPDT	file	PORTFOLIO-FILE	Update records
PORTDEL	file	PORTFOLIO-FILE	Delete records
PORTREAD	file	PORTFOLIO-FILE	Read records
DB2CONN	call	DB2ERR	Connection failure classification
DB2CMT	call	DB2ERR	Commit failure classification
DB2CMT	call	ERRPROC	Error reporting
DB2ERR	sql	ERRLOG	Insert error rows
DB2ERR	call	ERRPROC	Error reporting
DB2STAT	call	ERRPROC	Error reporting
ERRPROC	file	ERROR-LOG	Write diagnostics
AUDPROC	file	AUDIT-FILE	Write audit records
UTLMON00	file	DB2-STATS	Read collected statistics
UTLVAL00	file	POSITION-MASTER	Integrity checks

Source	Interface	Target	Detail
UTLVAL00	file	TRANSACTION-HISTORY	Reconciliation
TSTGEN00	file	PORTFOLIO-OUT	Generated test portfolios
TSTVAL00	file	TEST-REPORT	Test results
IBM RACF	authorizes	SECMGR	External security manager

Appendix-4

Layer Diagram and External Actors

Batch Processing layer	<i>JCL job step, non-CICS</i>
BCHCTL00, PRCSEQ00, RCVPRC00, CKPRST, HISTLD00, RPTPOS00, RPTAUD00, RPTSTA00, RTNCDE00, RTNANA00, POSUPDT	
Online Inquiry layer	<i>CICS TS, pseudo-conversational</i>
INQONLN, INQPORT, INQHIST, CURSMGR, SECMGR, ERRHNDL, DB2ONLN, DB2RECV	
Portfolio Maintenance layer	<i>JCL job step, non-CICS</i>
PORTMSTR, PORTADD, PORTUPDT, PORTDEL, PORTREAD, PORTVALD, PORTTRAN, PORTTEST	
Common Services layer	<i>Called subprogram</i>
DB2CONN, DB2CMT, DB2ERR, DB2STAT, ERRPROC, AUDPROC	
Utility layer	<i>JCL job step, non-CICS</i>
UTLMNT00, UTLMON00, UTLVAL00	
Test and Benchmark layer	<i>JCL job step, non-CICS</i>
TSTGEN00, TSTVAL00	

Data platforms: VSAM KSDS and sequential data sets; Db2 for z/OS database POSMVP.

External actor	Type	Interaction
Portfolio Analyst	person	CICS transaction PINQ via 3270 (mapset INQSET)
Operations	person	JCL job submission and batch control file
z/OS Scheduler	system	Batch job stream, process sequence file
IBM RACF	system	SECMGR authorization checks

Appendix-5

Roles and Responsibilities Matrix

Task	Maintainers	Architect	Reviewer	Translation Tooling
Issue and reissue this model	RS	A	RV	C
Layer allocation of a new program	RS	A	RV	-
Adding or changing a data store	RS	A	RV	C
Recording a deviation under Section 2	RS	RV	A	-
Consuming the model for translation	C	C	-	RS

RS - Responsible; A - Approver; RV - Reviewer; C - Consulted; - Not involved